

UNIT-II

Energy Storing Devices

LP3: Charging and discharging reactions in a reversible cell (Lead-acid battery) - Construction, working and applications

Secondary batteries

Secondary (rechargeable) batteries can be recharged by applying a reverse current, as the electrochemical reaction is reversible. The original active materials at the two electrodes can be reconstituted chemically and structurally by the application of an electrical potential between the electrodes to “inject” energy. These batteries can be discharged and recharged many times.

Applications:

These fall into two categories:

(a) The battery is used as an energy storage device. It is constantly connected to an energy source and charged by it. It can then release the stored energy whenever needed, e.g. in

- Car battery used to start engine
- Aircraft systems
- Standby power resources
- Emergency no-fail systems

(b) The battery is used as a primary battery would be but is then recharged instead of being disposed of, e.g. in

- Electric vehicles
- Mobile phones
- Cameras
- Power tools
- Toys
- Portable computers

Advantages:

- High power density
- High discharge rate
- Good low temperature performance

Disadvantages:

- Lower Energy density

- Poorer charge retention
- Safety issues
- Lack of standards
- High initial costs

The table below demonstrates the properties of various rechargeable batteries:

System	Nominal Cell Voltage (V)	Capacity (Wh/kg)	Advantages	Disadvantages	Applications
<u>Pb/Acid</u>	2.00	35	Low cost; good high and low-temperature operation	Low cycle life; low energy density; poor charge retention	Cars; lawn mowers; aircraft
Ni/Cd	1.20	30	Good physical durability; good charge retention; good cycle life	High cost; memory effect	Aircraft; emergency power applications
Ni/Fe	1.20	60	Good physical durability; long cycling and standing life	Low power and energy density; high self discharge; high cost	Stationary applications; fork lift trucks
Ni/Zn	1.60	27	High energy density; low cost; good low-temperature performance	Poor cycle life	Electric scooters/bikes; military vehicles
Zn/AgO	1.50	90	Highest energy density; low self discharge; high discharge rate	High cost; low cycle life; low performance at low temperatures	Military equipment eg torpedo propulsion, submarines
Cd/AgO	1.20	55	High energy density; low self discharge; Good cycle life	High cost; low performance at low temperatures	Portable power tools; satellites
Ni/H ₂	1.40	55	High energy density; good cycle life; can tolerate over charge	High initial cost; self discharge proportional to H ₂ pressure	Aerospace
Ag/H ₂	1.40	80	High energy density; good cycle life	High cost - limited to military and aerospace applications	Aerospace
<u>Li/Poly</u>	up to 4.2	135	High specific energy; good shelf life; moldable; non-volatile	High cost; expensive control methods needed for charge/discharge	Mobile phones

Lead acid battery or storage battery or reversible battery.

Construction

- A storage cell is one which can operate both as a voltaic cell and as an electrolytic cell.
- When it acts as a voltaic cell it supplies electrical energy.
- On recharging it acts as an electrolytic cell.
- Porous lead (lead sponge) as anode
- Lead-di-oxide as cathode
- Electrolyte: 20% Sulfuric acid

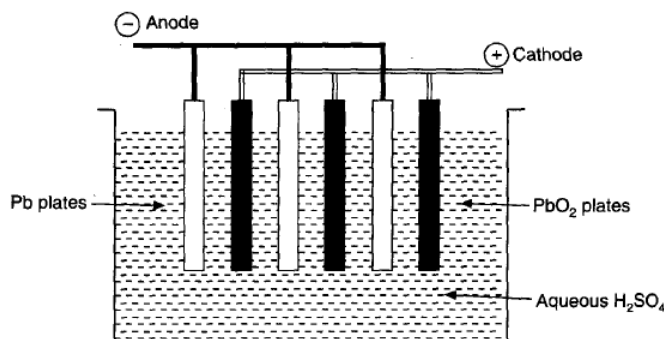
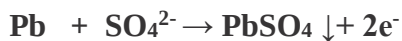


Fig. Commercially available Lead-acid battery

Working

Discharging (electrochemical cell):

At anode: Pb is oxidized to Pb^{2+} , which combines with SO_4^{2-} to give insoluble PbSO_4 .



At cathode: PbO_2 is reduced to Pb^{2+} that combines with SO_4^{2-} to give insoluble PbSO_4 .



Overall reaction:



From the above cell reaction it is clear that PbSO_4 is precipitated at both the electrodes and H_2SO_4 is consumed during discharging. If the density of sulphuric acid falls below 1.2 gm/ml,

the battery needs to be recharged. This can be done just by passing electricity in the opposite direction. *The best indication about the state of charge of the battery is Specific gravity of electrolyte.* The specific gravity decreases during the discharge of battery to a value near that of pure water (*This is because of the formation of PbSO_4 on discharging decreases the concentration of H_2SO_4*) and it increases during recharge (*This is because of conversion of PbSO_4 in to PbO_2 and Pb which increases the H_2SO_4 concentration*) . The battery is fully charged when specific gravity reaches its highest possible value.

Recharging (electrolytic cell):

On passing electricity exactly reverse reactions would take place. In other words, the electrical energy is converted into chemical energy.



As the lead-acid cell recharges:

- $\text{PbSO}_4(s)$ on each electrode is removed.
- Concentration of H^+ increases.
- pH of the electrolyte ($\text{H}_2\text{SO}_4(aq)$) decreases.
- *Overcharging a battery electrolyzes water, producing hydrogen gas and oxygen gas.*

Uses:

- Lead storage cell is used to supply current mainly in automobiles such as cars, buses, trucks, etc.,
- It is also used in gas engine ignition, telephone exchanges, hospitals, power stations, etc.,

Advantages:

It is rechargeable, portability, low cost and relatively constant potential

Disadvantages

- Relatively low cycle life.
- Big size and limited energy density.
- Hydrogen evolution in some designs could be dangerous.
- Use of conc. sulphuric acid is dangerous

Maintenance

- Avoid over discharging of the battery.
- Maintain the electrolyte at the proper level by adding water (whenever required).
- Keep the battery clean.

- Avoid overheating of the battery.
- Watering is the most neglected maintenance feature of flooded lead-acid batteries. As overcharging decreases water, we need to check it frequently. Less water creates oxidation in plates and decreases the lifespan of the battery. Add distilled or ionized water when needed.
- Check for the vents, they need to be perfected with rubber caps, often the rubber caps sticks with the holes too tightly.
- Recharge lead-acid batteries after each use. A long period without recharging provides sulfating in the plates.
- Do not freeze the battery or charge it more than 49-degree centigrade. In cold ambient batteries need to be fully charged as fully charge batteries safer than the empty batteries in respect of freezing.
- Do not deep discharge the battery less than 1.7V per cell.
- To store a lead acid battery, it needs to be completely charged then the electrolyte needs to be drained. Then the battery will become dry and can be stored for a long time period.
- *The most common type of water used in batteries is distilled water. Other types are deionized water and water from reverse osmosis. Ordinary tap water should not be used because it may contain an excessive amount of impurities that will degrade battery performance.*

Effect of temperature on Capacity and Efficiency

- The self-discharge rate increases as the temperature increases, due to the increased speed of chemical reactions within the battery. Irreversible damage (sulfation) occurs within a battery when stored at low states of charge (below 80% state of charge).
- Lead-acid batteries lose capacity in normal to cold weather by about 20% and in temperatures as low as -22°F(-30C) by about 50%.